GLOBAL WATER SECURITY & FLOOD RESILIENCE INITIATIVE

An Open Research Proposal for Integrated Water Management in Bangladesh

Proposed by: Md Abdul Aziz

Independent Researcher & Innovation Developer | Bangladesh

Project stage: Early-stage concept development; technical feasibility not yet established

Available resources: ■0 for experimental testing; seeking a partner or sponsor before testing

Purpose: Request for research collaboration and feasibility support

Research portfolio: https://mdabdulaziz30-web.github.io/md-abdul-aziz-research/

ORCID: https://orcid.org/0009-0004-8752-2107

This document describes a research concept, not a validated system or a claim of novelty.

1. Executive Summary

The Global Water Security & Flood Resilience Initiative is a proposed research effort to investigate whether selected water-management technologies can be combined in a practical, affordable, and environmentally responsible way for suitable communities in Bangladesh.

The concept considers four areas: atmospheric water harvesting, stormwater capture and storage, water treatment and reuse, and renewable-energy integration. These are existing fields of technology. The research question is whether selected approaches can be appropriately combined for a clearly defined local need, and whether the resulting system would offer measurable advantages over simpler alternatives.

No prototype has been tested under this initiative, and no performance, cost, novelty, or flood-reduction claim is being made at this stage. The project seeks a university or research-institution partner, technical mentor, laboratory access, and a sponsor willing to support an initial feasibility study.

2. Background and Rationale

Bangladesh faces varied water-management challenges that differ by location and season, including stormwater accumulation, flood exposure, water-quality concerns, and access to suitable water supplies. Different established technologies address different parts of these challenges.

Atmospheric water harvesting extracts moisture from air, but output and energy requirements depend strongly on climate and technology. Stormwater storage and drainage require site-specific hydrological assessment. Water treatment depends on the source and intended end use. Renewable-energy systems may support some operations, but their suitability and lifecycle costs must be evaluated.

These functions should not be assumed to work as one effective system simply because they can be combined conceptually. A structured feasibility study is needed before equipment is designed, purchased, or built.

3. Overall Aim and Objectives

Aim: To assess whether selected water-harvesting, stormwater-management, water-treatment, and energy technologies can address a defined water-security or flood-resilience need in a selected Bangladesh context.

• Review published research and existing commercial or demonstrated technologies.

• Identify one priority use case and assess local conditions, user needs, and existing alternatives.

• Compare potential configurations, including separate systems versus an integrated system.

• Define measurable criteria for water output, energy demand, water quality, stormwater performance, reliability, cost, and environmental impact, as applicable.

• Prepare a technically reviewed, costed plan for controlled component testing or a small pilot, subject to partner approval and funding.

4. Research Questions

• Which existing technologies are most relevant to the selected local problem?

• Under what climatic and site conditions could atmospheric water harvesting provide useful water, if included?

• What stormwater capture, drainage, or storage approach is appropriate for the selected site?

• What treatment steps and laboratory tests are required for the intended water use?

• Would renewable-energy integration improve performance after accounting for equipment, storage, maintenance, and energy demand?

• Does an integrated configuration offer a measurable advantage over simpler alternatives?

• What are the principal safety, environmental, maintenance, affordability, and implementation constraints?

5. Proposed Methodology

Phase 1 — Evidence review and problem definition

Review scientific literature, existing products, relevant standards, and local information. Identify prior art and potential intellectual-property issues before making novelty claims. Select one priority use case with advice from a technical partner.

Phase 2 — Site and baseline assessment

If a suitable site and local partner are available, document relevant conditions such as seasonal water availability, rainfall and runoff context, water-quality concerns, energy access, infrastructure, and user requirements. Establish a baseline for comparison.

Phase 3 — Technical comparison and feasibility design

Compare candidate technologies against measurable criteria. Evaluate whether each component is necessary, whether integration adds complexity, and whether an existing off-the-shelf solution would be more appropriate.

Phase 4 — Controlled testing, subject to funding

After technical review, safety planning, and funding are secured, test selected components at an appropriate scale. Use qualified laboratories for water-quality testing and competent technical supervision for electrical, thermal, structural, and other relevant hazards.

Phase 5 — Evaluation and reporting

Compare measured results with the baseline and stated objectives. Document methods, limitations, uncertainties, costs, unsuccessful approaches, and recommendations. Share results openly where permissions, safety, and partner agreements allow.

6. Evaluation Criteria

• Atmospheric water output, where relevant, in litres per day and energy consumed per litre.

• Stormwater volume captured, storage performance, and changes in site-specific runoff indicators.

• Water-quality results appropriate to the intended end use.

• Reliability, maintenance requirements, equipment lifetime assumptions, and operating costs.

• Capital cost and cost per unit of useful water or service, compared with alternatives.

• Energy source, greenhouse-gas implications, material use, and waste generated.

No claim of drinking-water safety will be made without suitable testing. No claim of flood prevention will be made without appropriate hydrological evidence. Atmospheric water harvesting will not be presented as a method for materially reversing global warming.

7. Partnership and Support Requested

The project currently has no funds available for experimental testing. Support is therefore sought in stages, beginning with non-cash collaboration where possible:

• A university or research organisation to review and, if appropriate, co-develop the research plan.

• A water, environmental, civil, hydrology, or energy specialist to provide technical mentorship.

• Access to relevant literature, data, laboratory services, or a suitable assessment site.

• A sponsor or grant partner to support the agreed feasibility work and, if justified, later controlled testing.

• Guidance on an appropriate legal, institutional, and financial structure for funded activity.

A detailed budget will be prepared with the selected technical partner after the research scope, testing requirements, and available facilities are established. Funding would be tied to an agreed work plan, transparent expenses, and defined deliverables.

8. Expected Outputs

• A documented review of relevant technologies and existing evidence.

• A defined use case and preliminary site or baseline assessment.

• A comparison of feasible technical options and their limitations.

• A risk and safety assessment.

• A costed feasibility and testing plan.

• A technical report describing findings, uncertainties, and a recommendation on whether to proceed.

These are proposed deliverables, not completed achievements. A prototype, field trial, or measurable community benefit is not guaranteed; each depends on the results of earlier phases.

9. Transparency and Responsible Research

The initiative will distinguish established evidence from hypotheses and planned work. It will not claim unverified novelty, demonstrated effectiveness, patentability, or social impact. Research design and safety requirements should be reviewed by qualified partners. Data, methods, limitations, and funding relationships will be documented as appropriate.

10. Long-Term Vision

If initial evidence supports further work, the research may be developed into a small, locally appropriate pilot and later assessed for adaptation to other locations. Replication would require fresh evaluation of climate, hydrology, water quality, infrastructure, cost, and community needs in each setting.

Contact

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Professional Outreach Email

Subject: Request for Research Partnership and Feasibility Support — Water Security & Flood Resilience in Bangladesh

Dear Research or Programme Team,

My name is Md Abdul Aziz, an independent researcher and innovation developer based in Bangladesh. I am seeking a research partner or sponsor for an early-stage concept titled **Global Water Security & Flood Resilience Initiative**.

The initiative proposes to investigate whether selected existing technologies for atmospheric water harvesting, stormwater capture and storage, water treatment and reuse, and renewable-energy integration could be appropriately applied to a clearly defined water-management challenge in Bangladesh.

This is a preliminary research concept, not a validated system. No prototype has been tested under this initiative, and I am not claiming that the proposed integration is novel or that it will necessarily reduce costs or flood risks. Establishing technical feasibility and comparing the concept with existing alternatives would be the first priority.

At present, I have no budget available for experimental testing. I am therefore seeking one or more of the following forms of support:

• Technical review and research mentorship;

• Collaboration with a university or qualified research team;

• Access to relevant literature, data, laboratory facilities, or a suitable study site;

• Assistance in developing a realistic methodology and itemised budget; and

• Sponsorship or guidance toward a suitable grant programme for an initial feasibility study.

I would be grateful if your team could advise whether the concept fits your research interests or whether there may be an appropriate researcher, programme, or funding opportunity to approach.

My research portfolio and ORCID profile are provided below for reference. I would be pleased to share the formal concept note and discuss a modest, evidence-based first phase.

Research portfolio: https://mdabdulaziz30-web.github.io/md-abdul-aziz-research/
ORCID: https://orcid.org/0009-0004-8752-2107

Thank you for your time and consideration. Any constructive feedback or direction toward an appropriate technical partner would be sincerely appreciated.

Yours faithfully,
Md Abdul Aziz
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Bangladesh

Before sending: Personalise the greeting and first sentence for each recipient. Confirm the organisation's current remit and eligibility rules. Attach this proposal and avoid implying that funding or a partnership has already been secured.